

ACML 100 Introduction to Chemistry (3-1-0)

Lecture outline

Module no.	Topic
1	Mole, Molarity, Dissociation constant of an acid, pH
2	Wave motion and light, line spectrum of atoms, Wave particle duality, Uncertainty principle
3	Wavefunctions, energy levels, particle in a box
4	Hydrogen atom, atomic orbitals, many electron atoms
5	Ionic bonds, Covalent bond, Lewis Structures, octate rule, Exceptions, VSEPR theory: predicting molecular geometries.
6	Valence bond theory and its limitations
7	Crystal field theory, High- and low-spin states, Spectrochemical series, Tetragonal distortion of octahedral complexes
8	Applications of crystal field theory, Color and magnetic properties of transition metal complexes
9	The Chemical Connection: Natural versus synthetic organic compounds
10	Organic acids and bases
11	Bonding in organic molecules: single, double and triple bonds
12	Stereochemistry in organic compounds, optical rotation, relative and absolute configuration
13	Substitution and elimination reactions of alkyl halides
14	How to determine organic reaction mechanism?
15	Structure and reactivity of organometallic compounds,
16	Importance of organometallic chemistry in metal purification and in catalysis
17	Rate of a reaction, order and molecularity of a reaction, Steady state approximation, pre-equilibrium.
18	Kinetics in determining mechanism of reactions.
19	Energy barrier for a reaction, catalysis, catalytic converters, Homogeneous catalysis, Enzyme catalysis

Recommended list of books:

1. Textbook of organic chemistry, R. T. Morrison and R. N. Boyd, Pearson Education.
2. Stereochemistry, David G. Morris, RSC, OUP
3. Organic chemistry, P. Y. Bruice, Pearson Education
4. Organic chemistry, Janice Smith, Mc Graw Hill
5. P. W. Atkins, J. de Paula, Physical Chemistry, Oxford
6. D. A. McQuarrie and J. D. Simon, Physical Chemistry, Viva
7. J. E. Huheey, E. L. Keiter, R. L. Keiter, O. K. Medhi, Inorganic Chemistry, Pearson
8. A. J. Elias, Basic Organometallic Chemistry, Universities Press

Course instructors:

1. Prof. Nidhi Jain
2. Prof. Shashank Deep
3. Prof. Subrata Kundu

Course evaluation policy:

One Minor: 30 marks (21-10-2024 to 25-10-2024)

One Major: 40 marks (16-12-2024 to 20-12-2024)

Assignments: 30 marks (six in number)